Year 10 Foundation

REVISION PACK

Shape and Space 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Q	Working	Answer
the exterior angle of a triangle		
1.1	$x = 47 + 68 = 115$	$x = 115°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
1.2	$x = 39 + 82 = 121$	$x = 121°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
1.3	$x = 55 + 64 = 119$	$x = 119°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
1.4	$x = 61 + 45 = 106$	$x = 106°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
1.5	$x = 33 + 90 = 123$	$x = 123°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
1.6	$x = 58 + 71 = 129$	$x = 129°$**. The exterior angle of a triangle equals the sum of the two interior opposite angles.**
angles in a quadrilateral		
1.7	$x = 360 - 96 - 81 - 92$	$x = 91°$
1.8	$x = 360 - 115 - 70 - 88$	$x = 87°$
1.9	$x = 360 - 78 - 108 - 95$	$x = 79°$
1.10	$x = 360 - 100 - 85 - 90$	$x = 85°$
1.11	$x = 360 - 120 - 65 - 80$	$x = 95°$

1.12	$x = 360 - 88 - 94 - 110$	$x = 68°$

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Q	Working	Answer
finding the hypotenuse		
2.1	$x^2 = 12^2 + 16^2 = 400$ $x = \sqrt{400} = 20$	x **= 20 cm**
2.2	$x^2 = 5^2 + 12^2 = 169$ $x = \sqrt{169} = 13$	x **= 13 cm**
2.3	$x^2 = 7^2 + 10^2 = 149$ $x = \sqrt{149} = 12.2066\ldots$	x **= 12.2 cm**
2.4	$x^2 = 8^2 + 15^2 = 289$ $x = \sqrt{289} = 17$	x **= 17 cm**
2.5	$x^2 = 9^2 + 11^2 = 202$ $x = \sqrt{202} = 14.2127\ldots$	x **= 14.2 cm**
2.6	$x^2 = 20^2 + 21^2 = 841$ $x = \sqrt{841} = 29$	x **= 29 cm**
compound shapes		
2.7	$\text{rectangle} = 14 \times 6 = 84$ $\text{triangle} = \frac{1}{2} \times 14 \times 4 = 28$ $\text{total} = 112$	**112 cm²**
2.8	$\text{rectangle} = 10 \times 7 = 70$ $\text{triangle} = \frac{1}{2} \times 10 \times 5 = 25$ $\text{total} = 95$	**95 cm²**
2.9	$\text{rectangle} = 16 \times 5 = 80$ $\text{triangle} = \frac{1}{2} \times 16 \times 3 = 24$ $\text{total} = 104$	**104 cm²**
2.10	$\text{rectangle} = 8 \times 6 = 48$ $\text{triangle} = \frac{1}{2} \times 8 \times 4 = 16$ $\text{total} = 64$	**64 cm²**

2.11	$\text{rectangle} = 18 \times 4 = 72$ $\text{triangle} = \frac{1}{2} \times 18 \times 6 = 54$ $\text{total} = 126$	**126 cm²**
2.12	$\text{rectangle} = 11 \times 9 = 99$ $\text{triangle} = \frac{1}{2} \times 11 \times 2 = 11$ $\text{total} = 110$	**110 cm²**
trapezium problems		
2.13	$\text{area} = \frac{1}{2}(9 + 15) \times 6 = 72$ $72 \div 20 = 3.6$ round up	**4 boxes**
2.14	$\text{area} = \frac{1}{2}(12 + 18) \times 8 = 120$ $120 \div 25 = 4.8$ round up	**5 boxes**
2.15	$\text{area} = \frac{1}{2}(7 + 13) \times 9 = 90$ $90 \div 25 = 3.6$ round up	**4 boxes**
2.16	$\text{area} = \frac{1}{2}(6 + 14) \times 5 = 50$ $50 \div 15 = 3.333$ round up	**4 boxes**
2.17	$\text{area} = \frac{1}{2}(11 + 17) \times 10 = 140$ $140 \div 30 = 4.667$ round up	**5 boxes**
2.18	$\text{area} = \frac{1}{2}(5 + 9) \times 8 = 56$ $56 \div 10 = 5.6$ round up	**6 boxes**
area of a circle		
2.19	$r = 12 \div 2 = 6$ $\pi \times 6^2 = 113.097\ldots$	**113 cm²**
2.20	$r = 18 \div 2 = 9$ $\pi \times 9^2 = 254.469\ldots$	**254 cm²**

2.21	$r = 9 \div 2 = 4.5$ $\pi \times 4.5^2 = 63.6173\ldots$	**63.6 cm²**
2.22	$r = 16 \div 2 = 8$ $\pi \times 8^2 = 201.062\ldots$	**201 cm²**
2.23	$r = 7 \div 2 = 3.5$ $\pi \times 3.5^2 = 38.4845\ldots$	**38.5 cm²**
2.24	$r = 22 \div 2 = 11$ $\pi \times 11^2 = 380.133\ldots$	**380 cm²**

Shape and Space 1 · Pythagoras' theorem (Lesson 5.3)

Q	Working	Answer
finding a shorter side		
3.1	$15^2 - 9^2 = 144$ $\sqrt{144}$	**12 cm**
3.2	$17^2 - 8^2 = 225$ $\sqrt{225}$	**15 cm**
3.3	$12^2 - 7^2 = 95$ $\sqrt{95} = 9.746\ldots$	**9.75 cm**
3.4	$9^2 - 4^2 = 65$ $\sqrt{65} = 8.062\ldots$	**8.06 cm**
3.5	$20^2 - 11^2 = 279$ $\sqrt{279} = 16.70\ldots$	**16.7 cm**
3.6	$8.5^2 - 4^2 = 56.25$ $\sqrt{56.25}$	**7.5 cm**

Shape and Space 1 · Circles (Lesson 5.5)

Q	Working	Answer
circumference		
3.7	$2 \times \pi \times 3 = 18.849\ldots$	**18.8 cm**
3.8	$\pi \times 12 = 37.699\ldots$	**37.7 cm**
3.9	$2 \times \pi \times 7.5 = 47.123\ldots$	**47.1 cm**
3.10	$\pi \times 25 = 78.539\ldots$	**78.5 cm**
3.11	$2 \times \pi \times 10 = 62.831\ldots$	**62.8 m**
3.12	$\pi \times 8.4 = 26.389\ldots$	**26.4 cm**

Shape and Space 1 · Angles (Lesson 5.1)

Q	Working	Answer
angles in an isosceles triangle		
3.13	$180 - 50 = 130$ $130 \div 2 = 65$	**65°**
3.14	$180 - 100 = 80$ $80 \div 2 = 40$	**40°**
3.15	$180 - 36 = 144$ $144 \div 2 = 72$	**72°**
3.16	Angle ACB = 55° $180 - 55 - 55 = 70$	**70°**
3.17	Angle XZY = 72° $180 - 72 - 72 = 36$	**36°**
3.18	$180 - 124 = 56$ $56 \div 2 = 28$	**28°**